PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

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Semester: V **Student ID: 25207002**
Class / Branch: SE B **Date of Performance: 19-10-26**
Subject: Agile Software Development & DevOps **Date of submit: 19-10-26**
Name of Instructor: Prof. Sameer Nanivadekar

Assignment No. 1

Q1 : Apply Agile principles and techniques to develop and prioritize User Stories with suitable Acceptance Criteria for any one application domain (E-commerce / Healthcare / College ERP / Banking / Food Delivery), and implement the Product Backlog using Jira. [LO1, L3]

Online Shopping Application

Search products Search

Available Products

Laptop - ₹45000 Add to Cart

Headphones - ₹1500 Add to Cart

Keyboard - ₹800 Add to Cart

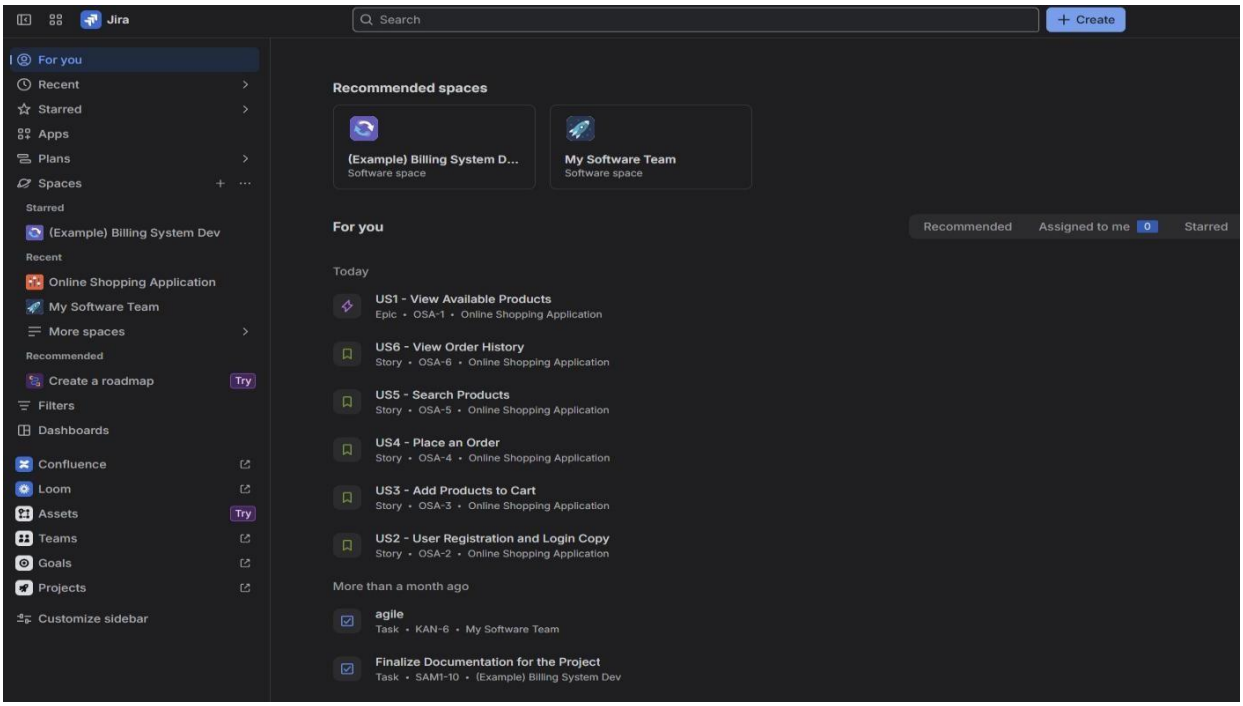
Mouse - ₹500 Add to Cart

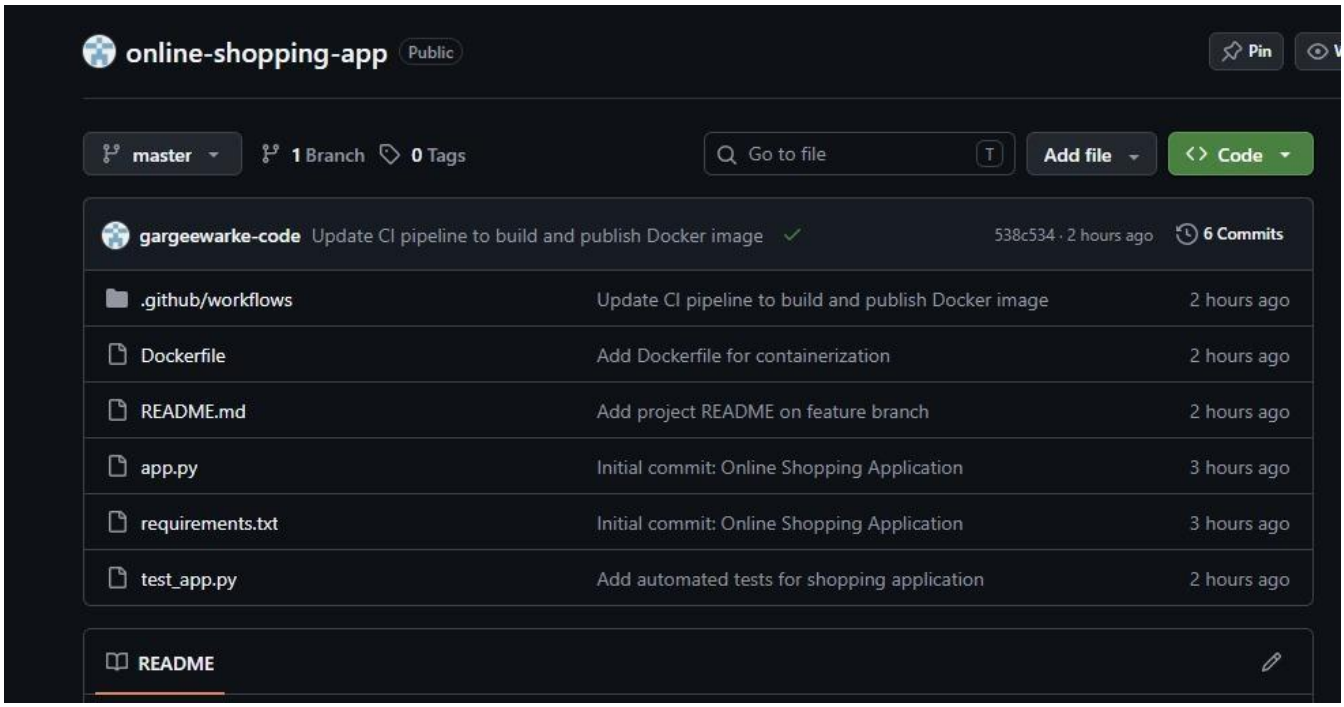
Shopping Cart

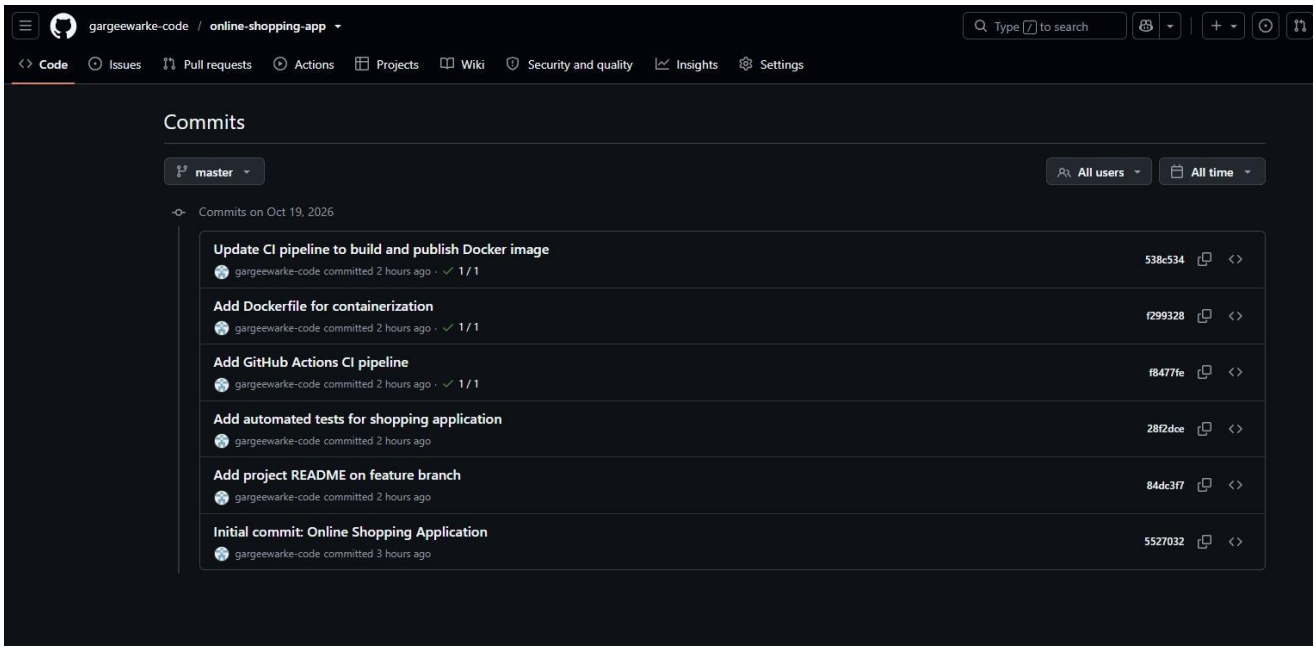
Your cart is empty.

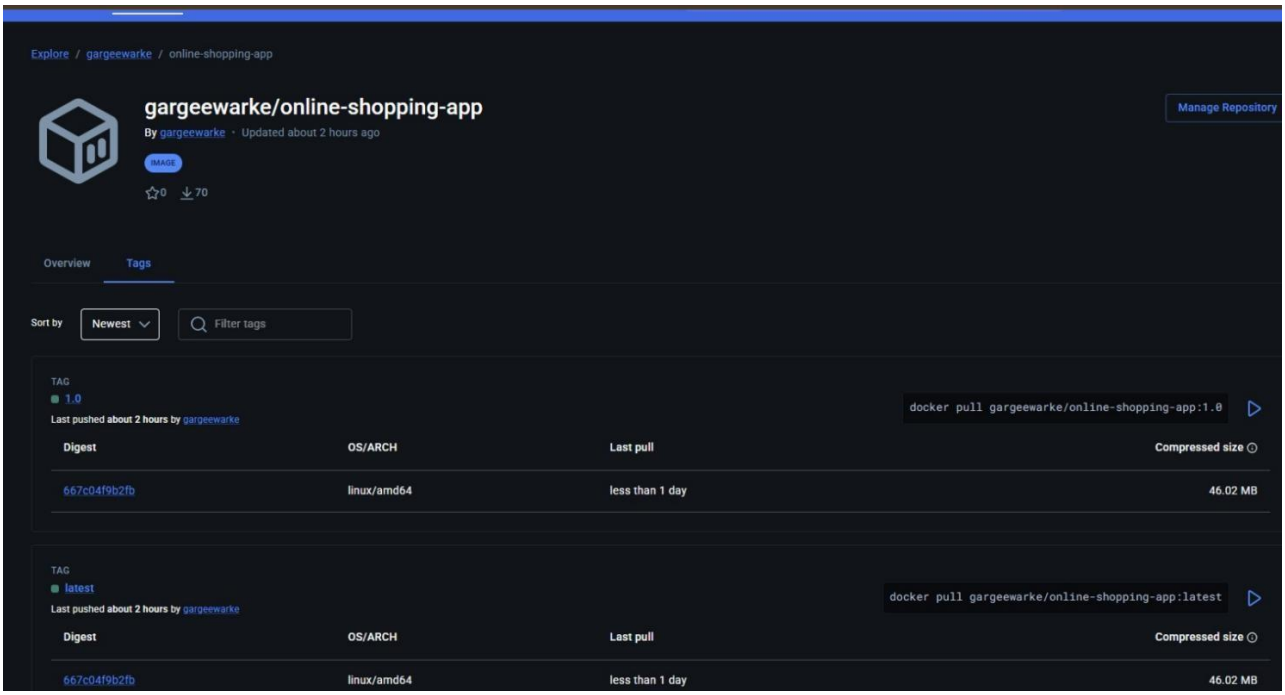
Total: ₹0

Place Order

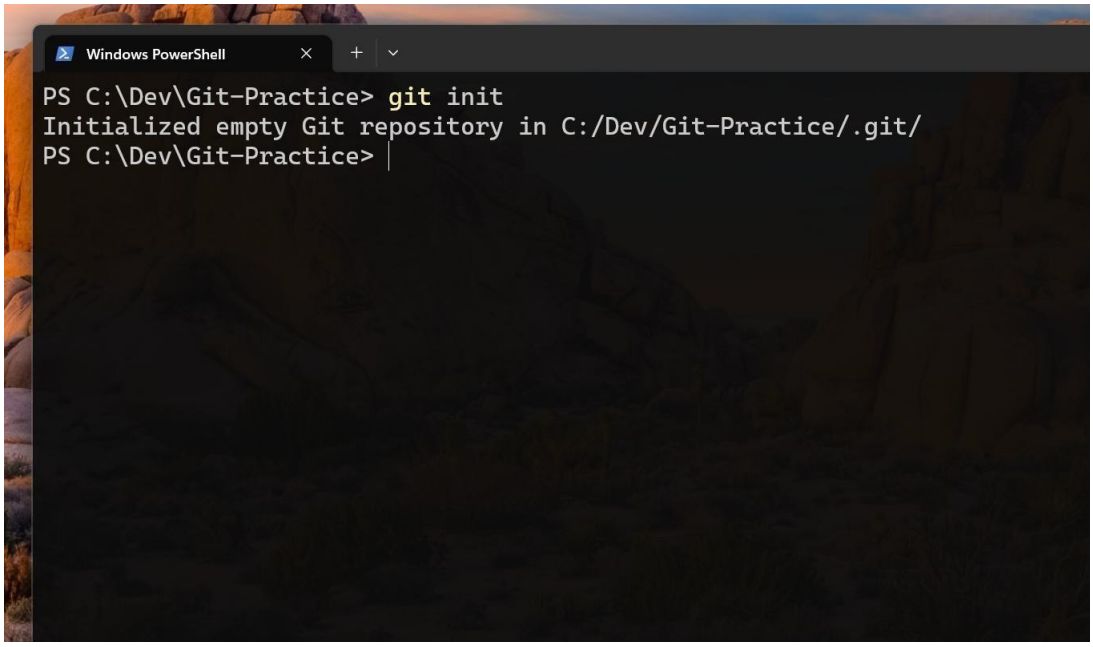

Jira
Search
Create
For you
Recent
Starred
Apps
Plans
Spaces
Starred
(Example) Billing System Dev
Recent
Online Shopping Application
My Software Team
More spaces
Recommended
Create a roadmap
Try
Filters
Dashboards
Confluence
Loom
Assets
Try
Teams
Goals
Projects
Customize sidebar
Recommended spaces
(Example) Billing System D...
Software space
My Software Team
Software space
For you
Recommended
Assigned to me 0
Starred
Today
US1 - View Available Products
Epic • OSA-1 • Online Shopping Application
US6 - View Order History
Story • OSA-6 • Online Shopping Application
US5 - Search Products
Story • OSA-5 • Online Shopping Application
US4 - Place an Order
Story • OSA-4 • Online Shopping Application
US3 - Add Products to Cart
Story • OSA-3 • Online Shopping Application
US2 - User Registration and Login Copy
Story • OSA-2 • Online Shopping Application
More than a month ago
agile
Task • KAN-6 • My Software Team
Finalize Documentation for the Project
Task • SAM1-10 • (Example) Billing System Dev


online-shopping-app
Public
Pin
master
1 Branch
0 Tags
Go to file
T
Add file
Code
gargeewarke-code Update CI pipeline to build and publish Docker image
538c534 · 2 hours ago
6 Commits
.github/workflows
Update CI pipeline to build and publish Docker image
2 hours ago
Dockerfile
Add Dockerfile for containerization
2 hours ago
README.md
Add project README on feature branch
2 hours ago
app.py
Initial commit: Online Shopping Application
3 hours ago
requirements.txt
Initial commit: Online Shopping Application
3 hours ago
test_app.py
Add automated tests for shopping application
2 hours ago
README


gargeewarke-code / online-shopping-app
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Commits
master
All users
All time
Commits on Oct 19, 2026
Update CI pipeline to build and publish Docker image
gargeewarke-code committed 2 hours ago · ✓ 1 / 1
538c534
Add Dockerfile for containerization
gargeewarke-code committed 2 hours ago · ✓ 1 / 1
f299328
Add GitHub Actions CI pipeline
gargeewarke-code committed 2 hours ago · ✓ 1 / 1
f8477fe
Add automated tests for shopping application
gargeewarke-code committed 2 hours ago
28f2dce
Add project README on feature branch
gargeewarke-code committed 2 hours ago
84dc3f7
Initial commit: Online Shopping Application
gargeewarke-code committed 3 hours ago
5527032


Explore / gargeewarke / online-shopping-app
gargeewarke/online-shopping-app
By gargeewarke · Updated about 2 hours ago
IMAGE
0
70
Manage Repository
Overview
Tags
Sort by
Newest
Filter tags
TAG
1.0
Last pushed about 2 hours by gargeewarke
docker pull gargeewarke/online-shopping-app:1.0
Digest
OS/ARCH
Last pull
Compressed size
667c04f9b2fb
linux/amd64
less than 1 day
46.02 MB
TAG
latest
Last pushed about 2 hours by gargeewarke
docker pull gargeewarke/online-shopping-app:latest
Digest
OS/ARCH
Last pull
Compressed size
667c04f9b2fb
linux/amd64
less than 1 day
46.02 MB

Q2 : Implement source-code management practices using Git and GitHub for the application developed in Assignment 1. Create a Git repository, perform commit management, create and merge branches, and collaborate with team members through GitHub. [LO2, L3]



Windows PowerShell — recreated terminal-style image

```
PS C:\OnlineShoppingApp> git init
Initialized empty Git repository in C:/OnlineShoppingApp/.git/
PS C:\OnlineShoppingApp> Get-ChildItem


    Directory: C:\OnlineShoppingApp


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        19-10-2026     17:27           2456 app.py
-a----        19-10-2026     17:28              5 requirements.txt

PS C:\OnlineShoppingApp> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Windows PowerShell — recreated terminal-style image

```
PS C:\OnlineShoppingApp> git add app.py requirements.txt
PS C:\OnlineShoppingApp> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   app.py
        new file:   requirements.txt

PS C:\OnlineShoppingApp> git commit -m "Initial commit: Online Shopping Application"
[master (root-commit) 5527032] Initial commit: Online Shopping Application
 2 files changed, 99 insertions(+)
 create mode 100644 app.py
 create mode 100644 requirements.txt
```

Windows PowerShell — recreated terminal-style image

```
PS C:\OnlineShoppingApp> git remote add origin https://github.com/gargeewarke-code/online-shopping-app.git
PS C:\OnlineShoppingApp> git remote -v
origin  https://github.com/gargeewarke-code/online-shopping-app.git (fetch)
origin  https://github.com/gargeewarke-code/online-shopping-app.git (push)
PS C:\OnlineShoppingApp> git push -u origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 22 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.21 KiB | 1.21 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gargeewarke-code/online-shopping-app.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Windows PowerShell — recreated terminal-style image

```
PS C:\OnlineShoppingApp> git switch -c feature-search
Switched to a new branch 'feature-search'
PS C:\OnlineShoppingApp> notepad README.md
PS C:\OnlineShoppingApp> git add README.md
PS C:\OnlineShoppingApp> git commit -m "Add project README on feature branch"
[feature-search 84dc3f7] Add project README on feature branch
 1 file changed, 44 insertions(+)
 create mode 100644 README.md
```

Windows PowerShell – recreated terminal-style image

```
PS C:\OnlineShoppingApp> git switch master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
PS C:\OnlineShoppingApp> git merge feature-search
Updating 5527032..84dc3f7
Fast-forward
 README.md | 44 ++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 44 insertions(+)
 create mode 100644 README.md
PS C:\OnlineShoppingApp> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 22 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 635 bytes | 317.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gargeewarke-code/online-shopping-app.git
   5527032..84dc3f7  master -> master
```

Windows PowerShell – recreated terminal-style image

```
PS C:\OnlineShoppingApp> python -m unittest -v
test_add_to_cart (test_app.ShoppingAppTests.test_add_to_cart) ... ok
test_home_page (test_app.ShoppingAppTests.test_home_page) ... ok
test_order_empty_cart (test_app.ShoppingAppTests.test_order_empty_cart) ... ok
test_search_product (test_app.ShoppingAppTests.test_search_product) ... ok

----------------------------------------------------------------------
Ran 4 tests in 0.064s

OK
PS C:\OnlineShoppingApp> git commit -m "Update CI pipeline to build and publish Docker image"
[master 538c534] Update CI pipeline to build and publish Docker image
 1 file changed, 27 insertions(+), 5 deletions(-)
PS C:\OnlineShoppingApp> git push origin master
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 22 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 886 bytes | 443.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/gargeewarke-code/online-shopping-app.git
   f299328..538c534  master -> master
```

Q3: Apply CI/CD practices to the application and Git repository developed in Assignments 1 and 2 by configuring and implementing an automated CI/CD pipeline using Jenkins or GitHub Actions to build, test, package, and deploy the application. [LO3, L3]

Code Blame 50 lines (40 loc) · 1.24 KB

```yaml

name: Online Shopping CI/CD Pipeline

on:
  push:
    branches: ["master"]
  pull_request:
    branches: ["master"]

jobs:
  test-and-publish:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"

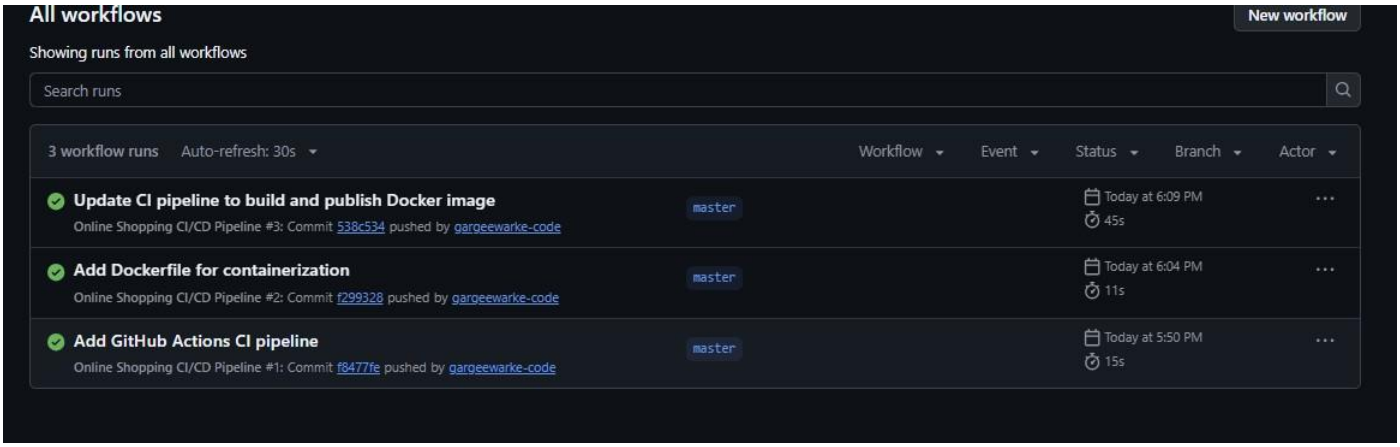
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt

      - name: Run automated tests
        run: python -m unittest -v

      - name: Set up Docker Buildx
        if: github.event_name == 'push'
        uses: docker/setup-buildx-action@v3

      - name: Log in to Docker Hub
        if: github.event_name == 'push'
        uses: docker/login-action@v3
        with:
          username: ${{ secrets.DOCKERHUB_USERNAME }}
          password: ${{ secrets.DOCKERHUB_TOKEN }}

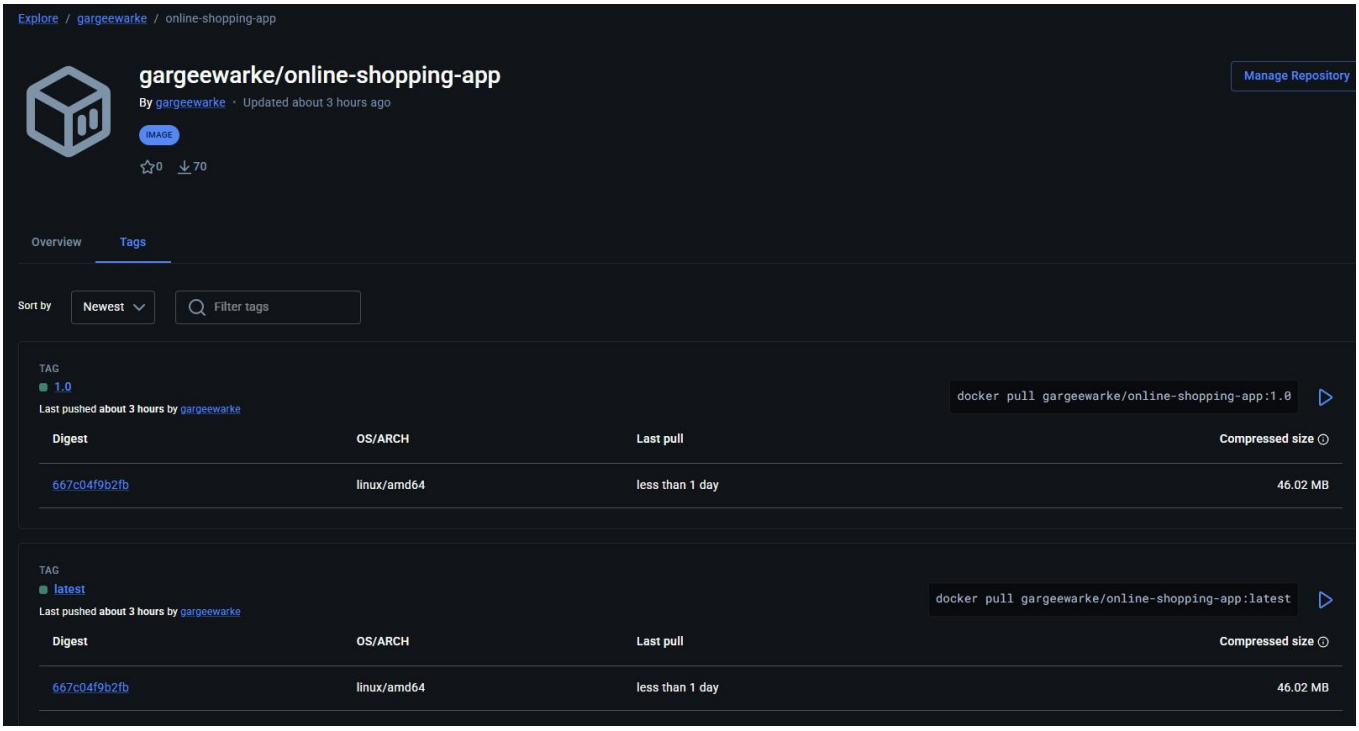
      - name: Build and push Docker image
        if: github.event_name == 'push'
        uses: docker/build-push-action@v6
        with:
          context: .
          push: true
          tags: |
```



```
PS C:\OnlineShoppingApp> cd C:\OnlineShoppingApp
PS C:\OnlineShoppingApp> python -m unittest -v
test_add_to_cart (test_app.ShoppingAppTests.test_add_to_cart) ... ok
test_home_page (test_app.ShoppingAppTests.test_home_page) ... ok
test_order_empty_cart (test_app.ShoppingAppTests.test_order_empty_cart) ... ok
test_search_product (test_app.ShoppingAppTests.test_search_product) ... ok

----------------------------------------------------------------------
Ran 4 tests in 0.064s

OK
PS C:\OnlineShoppingApp>
```



```
PS C:\OnlineShoppingApp> python app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.0.118:5000
Press CTRL+C to quit
127.0.0.1 - - [19/Oct/2026 17:29:47] "GET /?utm_source=chatgpt.com HTTP/1.1" 200 -
127.0.0.1 - - [19/Oct/2026 17:29:47] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [19/Oct/2026 17:30:11] "GET /add/1 HTTP/1.1" 302 -
127.0.0.1 - - [19/Oct/2026 17:30:11] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [19/Oct/2026 17:30:13] "GET /add/2 HTTP/1.1" 302 -
127.0.0.1 - - [19/Oct/2026 17:30:13] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [19/Oct/2026 17:30:15] "GET /add/3 HTTP/1.1" 302 -
127.0.0.1 - - [19/Oct/2026 17:30:15] "GET / HTTP/1.1" 200 -
```